Western Philosophical Thought

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In this paper, I present a theorem that might help to explain the language biases in western philosphical thought, of the search for universals in metaphysics and excluded middle in logic.

Western philosophical thought is an extremely complex phenomena. It spans millennia, huge areas of geographic influence and millions of people who have contributed with their own ideas. Therefore, it might seem impossible at first, to suggest that there is something about western philosophical thought can be thought of as central, without being perceived as intellectual dishonest. Yet, thinkers like Kent Palmer (PhD) have spent many years looking for ways to explain western philosophical thought as a whole. The framework of philosophy needed to provide an account of all the connections in western philosophical thought is complex, but at the end of the day, the argument boils down due to the influence of *Syllogism*, a retraction from natural language into constrained sentences and a set of rules of valid inference. Multiple versions existed.

Path Semantics is a new subfield of Semiotics that studies how people use symbols. This field is related to Linguistics, Pragmatics, Logic, Mathematics and Theoretical Physics, but takes a different approach. The basic idea is that use of symbols can be modeled in formal systems and these models have logical consequences that can help to explain techniques used in intellectual disciplines.

In Path Semantics, a Seshatic-Platonic language bias cycle is a cultural trend where people alternate between different ways of crediting knowledge. Seshatism credits knowledge by causality and Platonism credits knowledge by abstraction. Parts of human history have been obscured by religious propaganda, which makes it difficult to understand what religious thought was in the past. Seshatic-Platonic language bias cycles can help to reconstruct these changes and connections to philosophy. Seshatic-Platonic language bias cycles might be used in the future to start a galactic civilization, that operate on extremely large scales. The motivation to test the theory against human history, is to explain phenomena that can not be easily simulated by computers using current technology.

A lot of work through study of language biases in Path Semantics has been put into figuring out schematic chronology of 2nd century Early Christianity and its relation to practice of medicine. This is needed to fill some gaps that were previously obscured by religious propaganda, in order to provide a better perspective of Seshatic-Platonic language bias cycles overall from current available evidence. However, there will be many years still before this work reaches a high level of quality.

While much still remains to be done, there is enough evidence at the moment to suggest that religious thought, developed during 2nd century Early Christianity, is under influence of philosophy. Particularly the concept of the Holy Trinity, can be thought of as part inspired by 2nd century breakthroughs in medicine and part by latent psychological reasoning about semantics in human brains that take the form of partial equivalence as the mechanism to reach for infinite-valued logic. This hypothesis is supported by literary analysis of Early Christian writings, who use a lot of Roman satire to mimic contemporary elite literature, which establishes a link between Jesus Christ as a literary character with Simon Magus, a savior figure in a competing mystery cult. Jesus Christ and Simon Magus in the 2nd century function as school brands for education within medicine, which explains why Neo-Platonic ideas have high influence on the theology behind the Holy Trinity.

In this context, the following theorem is relevant (for proof, see **Appendix A**):

$$\text{true} \to \text{all}(\text{excm}(a)\hat{}\text{true}) == \text{all}(!!a \to (b \mid a \mid c) == ((b \mid a) \,\&\, (a \mid c)))$$

The theorem establishes a link between the law of the excluded middle, which one can add to constructive logic to get classical logic, and a certain desirable compactification technique in a potential lost philosophical text, which would describe a study of a pre-grammar for initial objects in the straightedge and compass constructions advocated by ancient geometry in Euclid's *Elements*.

In ancient astronomy, the planet Venus is of major interest, because its orbit close to Earth produces pertubations in a geocentric model that ancient astronomers explained using the technique of epicycles. This is part of the reason that in ancient theology, the planet Venus is associated with various savior figures, because these savior figures function as brands for mystery cults that provide opportunities for children to get an education within medicine. In the epicycle technique, circles play a major significance, which ratios inspired a form of numerology that in Early Christianity during the 2nd century was linked to calendars of religious rituals and agricultural practice. Here, much of the focus is on the numbers 12 and 13 respectively, which describe the months in calendars. The author of Revelation might have learned that 12 times 12 is 144, which is the smallest number with 13 divisors. Divisors is a mathematical concept that was popular in the ancient world, prior to the shift toward primes as the major topic in number theory.

This establishes a link between the focus on circles in geometry and practice of medicine. The dominant text that influenced western philosophical thought about circles, was Euclid's *Elements*. Therefore, it is likely that the *Elements* had some influence on western philosphical thought as a whole. While this hypothesis is supported by multiple lines of evidence, due to the popularity of this text, there are two features of western philosophical thought that were previously unexplained:

1. The search for universals in metaphysics.
2. The law of excluded middle in logic.

Up until now, scholars have thought that Plato is an influential figure in philosophy that motivated the search for universals in metaphysics. Aristotle, Plato's student, is seen as an influential figure in philosophy that motivated the law of excluded middle in logic. However, the newly discovered theorem might help to establish a link of these two ideas to Euclid's *Elements*, in a way that also shines some light on the development of Early Christian theology during the 2nd century.

Plato's view of metaphysics is based on previous philosophers, like Parmenides and Heraclitus, who in turn are influenced by the language of mystery cults. When considering that mystery cults are closely related to education within medicine, which in turn applies philosophical training to teach students critical thinking skills, it might not be surprising if Plato echoes ideas from contemporary intellectual environments. Some ideas that to modern people might seem original in Plato's texts, might have been widely accepted prior to his work. This suggests that one might take a better look at what precisely is meant by universal in metaphysics, what kind of arguments that Plato makes when interpreting them through the language of mystery cults, and how this view fell out of popularity during the early 18th century through Amo's dissertation on the mind-body problem.

In the language of mystery cults, used by people who often associated the planet Venus with various savior figures, the fundamental property of existence is linked to the idea of a communion with the divine. It means, these people have a cosmology where earthly existence is reflected by hidden and sacred laws in the sky. Associating various savior figures with the planet Venus is a manifestation of one such belief. It is important to not overlook these superstitious beliefs from a modern lens, because they provide a perspective into the mind of ancient people. Certain numbers, ratios and patterns in nature, where thought to reflect human experiences of sounds, e.g. in music, colors, e.g. in art, and altered states of consciousness, e.g. through use of entheogens in religious rituals.

Basically, many ancient people viewed communion with the divine as the explanation for existence.

The practice in mystery cults is based on the idea to bring all aspects of human experience that might open up a space for communion with the divine, into one ritual. Music, art and entheogens were brought together to represent universals of human experience. This ritual was kept secret and only accessible to the initiated, to protect their cultural identity, social networks and knowledge of medicine, from competing mystery cults. This is where savior figures come in, because they are literary devices used in advertisments to attract people toward a particular mystery cult, advocated by its followers. Ancient people spread religious beliefs this way, to expand their own influence.

A circle, while being a conceptual tool used in geometry and in calculations using the epicycle technique in astronomy, is also a cultural symbol of a closely tied social network. The circle represents the inner group of mystery cult practitioners, who decide the format of their religious rituals, calendars and the content of their religious texts.

When people in the inner circle of some mystery cult disagree, some members might break out and build their own inner circle to increase their own influence further. This can lead to heated debates and hostily between competing mystery cults. In a single family in the ancient world, some members might belong to one mystery cult, while the others might belong to another mystery cult, because the family previously belonged to one mystery cult in the past that split up into new ones. The cultural significance of association the planet Venus with various savior figures, resulted in disputes over which savior figure was the correct one, but disagreements over how to establish such beliefs based on their available evidence. In response to such critique, people often invented stories, using contemporary elite literature as inspiration, combining it with ideas extracted from historical works, to attempt convince their followers that their beliefs were rooted in some historical events.

During the 2nd century, Early Christians were inspired by contemporary Roman satire, which was seen as the prestige of Roman literal achivement, that was not inherited from Ancient Greece. Satire also played a role in education of philosophy, because teachers designed puzzles in the form of anecdotes, which the students analyzed and took apart into their satirical building blocks, which they recognized as solved by finding a core of some logical statement. With other words, the hidden logical structure inside the satirical literary elements of an anecdote used in philosophy, is not taught directly as an immediate teaching moment in the sense of modern education, but reflected in the practice of medicine through the language of mystery cults. Roman satire was seen as, not just as social critique, but a way of elevating understanding of the whole cosmos.

The significance of the concept of universals in metaphysics can not be understated, because it is seen as the very mechanism that ancient people thought about existence overall. Sounds, colors and altered states of consciousness are the three human experiences that are given holistic interpretation and attributed spiritual and psychological meaning.

Universals in metaphysics were tied conceptually to human experience, but the character of this concept was based on the perspective of reasoning, or thinking, simultaneously by giving the concept an philosophical account for overall existence. This view was popular in western philosophical thought up until early to mid 18th century, when Amo, an African philosopher working in Germany, used his knowledge of medicine to argue that human experiences, such as seeing a color, happened in the body and not in the Cartesian mind of reasoning. At the time, Germany became more racist and advocates of education for all people were forced into silence. Therefore, philosophers who used Amo's dissertation were seen as potential threats to status quo. It became a practice to not quote Amo directly and not quote philsophers who quoted Amo. Philosophers like Kant did not like to be associated with earlier philosophers, like Wolff, because Wolff's open minded views on education of women and people of color. Thus, Kant claimed that his ideas were new and not based on this earlier tradition of thought. However, persuading arguments by Amo that these previously universals of human experience, were not universals, influenced Kant's ideas.

Philosophy after the 18th century departs from the prior association of universals in metaphysics to human experience and looks for a pure notion of reasoning, which would later result in the development of formal logic. In modern times, this development has been overlooked because people read Plato today without the influence of Amo's dissertation. Therefore, many philosophers today still consider some human experiences, such as sounds and colors, as universals in some sense. The mind-body problem is seen as an attempt to explain how these human experiences map onto a scientific description of the physical processes in the human brain. However, this is based on the intuition that a univeral needs to be some form of unification between different descriptions of reality that have very different language biases. Amo's disseration focused on, not on unification, but to separate these concepts. It was in the complete opposite direction.

In ancient mystery cults, the mind-body problem can be understood as explained by a communion with the divine through performing a sacred religious ritual. This ritual is symbolic, but also a personal journey, which is tied with the social group that the person belongs to. Therefore, there is no clear separation between individual existence and the existence of social groups. Again, this idea is associated with overall cosmic harmony. From a modern scientific lens, this idea seems simply to be nonsense, but packaged into a cultural setting that served important functions through history.

In Path Semantics however, use of symbols is in itself rich of mathematical structure. One can argue from a path semantical perspective, that what was seen as an education in philosophy in the ancient world, reflected the social constraints and pragmatic practice of medicine. Instead of seeing nonsense as something that must be explained away using scientific terminology, nonsense can also be embraced and used constructively. This is motivated by the example of the mind-body problem, because no matter how much people tried to make sense of it, no satisfactory solution has been found, perhaps because people do not study language biases and their cycles in complex systems.

Our current technology does not allow us to study language biases directly through simulated experiments, which makes it hard to evaluate their exact properties. The AdvancedResearch community constructs simple models that demonstrate aspect of a few language biases, so they can be compared to other models with different language biases. This methodology is flawed, but it has provided some insights into the nature of consciousness, when interpreted through the lens of theoretical physics. However, at the moment there is still much to be done before one can argue that consciousness in relation to existence is fully understood. There are also some arguments, based on extreme observer selection effects due to Eternal Cosmic Inflation, predicted by most theoretical models in cosmology, that the relation between consciousness and existence might never be fully understood, due to existence being uniquely locating an individual person within the universe, while continuous and rapid expansion of the overall universe might prevent any observable evidence to be used to distinguish one observer from another similar observer, symbolically.

Despite departure during the 18th century from the notion of universals in the sense of the language of mystery cults, that was popular in the ancient world, modern research on the mind-body problem is still theoretically linked to the cosmology of the overall universe in relation to the individual.

Now, a universal in a formal sense, can in Path Semantics be thought of as an axiom of the form:

all(a^true) where `a` is some expression

For example, in classical logic, the excluded middle is an axiom that expresses a universal property:

all((a | !a)^true) also written as all(excm(a)^true)

It means, every proposition is either true or false. This is a language bias that does not always hold.

It is important to recognize that a system of reasoning that has language biases, for example a binary computer using excluded middle, can also function as language tool for general reasoning. Just because some system has language bias, might not limit the applications of the system itself. Language biases is a phenomena that influences how people think and see the world around them in general, because people use symbols whenever they represent some knowledge to themselves or share this knowledge with other people. Language biases are not important philosophically to give an account of all of existence, precisely because in strict philosophical terms, it is just a biased perspective. A person can be fully immersed in some language bias and entirely convinced that this way of thinking is the proper one, without these views being shared with another person, fully immersed in the dual language bias. Two such people using different language biases as basis for their own thinking, can disagree due to the language biases themselves and never find a common ground. Recognizing language biases is important in philosophy to get unstuck from such debates.

In Path Semantics, there are two fundamental language biases: Seshatism and Platonism. From these language biases, one can construct higher dualities using Joker Calculus, which has normalization rules in two variants, Open and Closed, that reflect common human psychology.

A universal from the perspective of Path Semantics in a formal sense, is therefore an axiom that a person might use, depending on what kind of mathematical language they are designing.

This is part of the reason that Path Semantics is simply the study of how people use symbols. A path semanticist is not judging use of symbols based on ontological or metaphysical grounds, but merely as design choices. When studying Seshatic-Platonic language bias cycles, one seeks to uncover the underlying structure of a flow of ideas through history, but such flows are not given existential priority. From a path semantical perspective, the theory is merely descriptive of such flows of ideas.

A such path semantical perspective is needed to embrace the ideals of ancient philosophical thought, to see these views through the lens of how people used symbols at that time, instead of interpreting these views through a modern scientific lens only. The construction of this lens is heavily based on modern science, but the goal is to get a glimpse of what ancient people might have been thinking about.

In 2nd century Early Christianity, there is a lot of debate whether the divine should be understood as a trinity or a duality. A trinity in theology is a system of 3 deities that are unified in some sense. Similarly, a duality in theology is a system of 2 deities that are unified in some sense. While this seems somewhat arbitrarily, when one considers that ancient people might have thought about language biases themselves as signs of the divine and supernatural phenomena, it is natural to investigate whether Early Christians thought about 2 deities. Indeed, for example in the gospel of John, one can find traces of Logos and Zoe in the beginning. In later traditions, Logos in the first verse was given much attention, while Zoe in the second verse was ignored. This is because, in Late Christianity, Logos is seen as one of the embodied organs of the Platonic tripartite soul of divinity. To pay attention to Zoe, would conflict with this view. This is why Zoe gets ignored.

When Early Christians write their texts during the 2nd century, they do not operate with a fixed theology where the divine is seen as either a trinity or a duality. Instead, these ideas were contemporary and wide spread in philosophical circles. So, while there were indeed disputes over whether to start with a trinity or duality in metaphysics, the perspective of these disputes were mostly theoretical and not seen as having much practical significance. Therefore, most religious thought at the time was tolerant toward both perspectives. Now, the people who wrote texts were often arguing for one side, which does not reflect the social dynamics of tolerance, but more like representing a thesis why one side should be correct. This cultural tolerance changed when Galen performed public vivisection experiments on pigs. Which he claimed demonstrated a tripartite soul.

Since Galen was the physician of several Roman emperors and a prolific author, his views on the tripartite soul, based on a Platonic perspective, became a doctrine in the later practice of medicine, where the tripartite soul was linked to three body organs: The brain, the heart and the liver. Different human experiences, which were viewed as universals, were thought to be located within these organs. The nerves connecting these body organs, when separated by surgery in living pigs, which is vivisection and considered animal abuse by modern ethical norms, the behavior of the pigs, before they died, was changed in a such way that to ancient people, they thought it gave them insights into the locations of their conceptual universals, mapped onto the three organs of the body.

This view of the tripartite soul as having physical manifested evidence, had a great influence on western philosophical thought, because in the dispute over a divine trinity and a divine duality, which were previously argued purely on metaphysical grounds, was thought as being settled, due to the experimental nature of Galen's argument. Ancient people put much more emphasis on embodied divinity, usually seeing deities as invisible super-humans, protecting cities using temples as their homes. Furniture in temples were often over-sized for divine symbolism. Galen's argument was a precursor to the scientific method. It did not prove that a divine trinity was correct, but people thought that it proved a divine duality was incorrect. This is why they settled with a divine trinity.

Still, while this hypothesis explains how theology developed the idea of the Holy Trinity during the 2nd century, one must also explain why Early Christians thought about divine trinity and divine duality in the first place. One approach is that this is simply arbitrary. Ancient people were often superstitious, therefore, one can argue that whatever reasons they had to disagree over such views, were not rational. However, this approach fails when considering that very smart people were deeply involved in these arguments, who also presented these arguments for or against positions on the existence and nature of universals in metaphysics. The missing link here is a metaphysical perspective, possibly a lost text, that argues for some solution that results in a trinity or a duality.

The relevant theorem here establishes a such link, between a universal notion of the excluded middle, and a universal notion of unification of trinity with duality. Now, one might dismiss this idea as absurd, because, obviously, three and two are in contradiction. How can they be unified? The argument here is that if a such unification could be proved, in a sense that convinced ancient people, the search for further arguments that claimed trinity or duality as a favored priority in metaphysics, makes sense. It is not a contradiction between trinity and duality that motives the disputes. On the contrary, it is the idea that these two might be unified in some sense, so figuring out which one of them that comes first, presents an interesting puzzle for intellectuals to solve.

This is where Euclid's *Elements* comes in. In this text, there are some postulates that describe valid constructions in geometry. These constructions are based on available instruments for drawing that ancient people could build. Furthermore, at larger scales of governance, such as landscape surveys needed to decide properties and to plan for war, the drawing instruments used for constructions in geometry have their analogues, which inference provide valuable knowledge to exploit society.

The basic principle of power in society, is that a ruling class of people manages to exploit people who are not inside their own inner social circle. One can think about ancient mystery cults as satirical playgrounds for mimicry of the actual social power structure. Like the drawing instruments used by Euclid's *Elements* have analogues in landscape surveys, the inner circles of mystery cults function socially in a way that reflects the inner circle of the ruling class.

One can also see this phenomena in the language of what Early Christians considered holy scripture. The savior figure and the pantheon of divine characters use the same terminology that one uses to describe the power structure of the ruling class. For example, Jesus Christ is seen as future king, which will rule over a future kingdom. It is a training ground for kids to pretend they will rule.

Physicians like Galen had great influence over theology, precisely because ancient religion is largely a form of education for kids, so they can experiment safely with the ideas without harming their own group. Later, when these kids grow up and become adults, they apply the knowledge they learned, to build economies and wage war on enemies. The language of medicine flows in one direction, from the elite, to kids, who gets indoctrinated into a religious belief system designed to ensure their long term loyalty to some specific group of interest. Philosophy in this context is a branch of medicine, used to teach kids critical thinking skills, where religious stories functioned as requirement to make kids interested in studying philosophy. Religion and philosophy in the ancient world are not in conflict culturally, but enforce and support each other. However, when you read a single text, you get the impression that philosophers were against religious superstition, precisely because they are arguing for their own views and against competing views. The missing context is the social setting and the power dynamics due to practice of medicine.

Now, keep in mind that the circle is the fundamental geometrical shape, used in geometry and astronomy, which also serves as a metaphor for the closely tied inner circle of a mystery cult. This inner circle is mimicking the ruling class of society. Not everyone in the inner circle of a mystery cult have power. What they have in common, is that they want more power. Therefore, a circle in this sense symbolizes the desire for power. That is why the circle is also used symbolically in religious rituals. Ancient people perceive religious rituals as ways to get in contact with divine powers, from which they believe they can alter and influence the material world.

Jesus Christ during the 2nd century is not a deity in the sense what people think of as a deity in modern religion. Primarily, Jesus Christ was a savior figure, a brand for a type of school, where students learned to read and write by copying stories about this character. The copies made by students as exercises circulated in the cultural environment around the school, which people read aloud to others who could not read. Those who found things they did not like in these stories, often made their own versions, to spread their own views. The highest authority of such stories was the school itself, but people in the proximity of the school could create their own texts for entertainment. When people wanted to know the “correct” version of a story, they were referred to the school and its teachers. However, inside the school, the main purpose of these stories are not about what they contain, but to provide exercises so that the students can learn to read and write.

Think of this from the perspective of a parent: A kid who learns to read and write, might come up with some ideas that they put on paper. A parent can read this text for other grown ups, out of pride and amusement, without attributing the text some existential significance. Once in a while, there is a kid that gets an idea that pique the interest of grown ups. In return, the kid receive appraisal. This causes the kid to try repeat the performance. Over time, as a kid develops this skill, this becomes a personal identity, which in the end might result in a career as a teacher, a scribe, or as a physician.

The development of ideas in religious thought are intrinsically linked to the development of intellectual skills in children. When these children grow up, they can still find pleasure in these activities, even though it is not their primary function in society or of economic importance. Childish beliefs were often idealized, as source of genuine truths, partly because of this kind of unfiltered creativity. However, such beliefs and ideas were not doctrines. They were expressions of personal opinions and cultural identity. Doctrines were mostly limited to practice of medicine.

Now, when a religious text gets copied widely, it gains a cultural significance that was not in the context of the use of the original text. This is how authority is attributed to texts themselves, even the author might not agree with a such use. A joke can be interpreted later on as a serious point. Sometimes, wars were waged, using disagreements over such interpretations as excuse. This is also a form of common abuse by the elites, which often used ridiculous arguments to go to war, because this causes the harm they do to their enemies, feel even more pointless and wasteful, on purpose.

After many wars have been waged in the name of ridiculous arguments, people start to believe that there must be something intrinsic of value in such arguments. For example, arguments that are used to oppress some minority, often come from a bloody history and carry baggage precisely because of the wars and conflicts in the past. This was how the Holy Trinity became a doctrine, because it became a symbol which the majority decided to use, to identify their loyalty to their own group, while opposing views became symbols of minorities that the majority saw as potential enemies.

There is no need to explain the Holy Trinity as something somebody in history once genuinely believed. Galen's influence in the 2nd century and later in medicine, is sufficient to explain the concept of the Holy Trinity as a reaction to arguments and disputes on metaphysical grounds, from which the Euclid's *Elements* might have played a significant role. To explain this influence, one must take a closer look at initial object in geometry.

Euclid's *Elements* is based on the principle that one can provide a small number of axioms or postulates and build a large and complex theory out of it. The axioms describe the valid constructions, such that the whole result can be divided into simple steps. However, the theory does not determine a starting point for doing geometry.

For example, there is no axiom in the theory that claims one has a point on a plane to start constructing something. What happens in practice is that a person first draws a point, maybe a line, and the rest of the construction follows the axioms. These are initial objects in geometry.

Initial objects can have arbitrary complexity and might not even be constructible using the axioms of geometry. For example, a convex shape, as an idea in general, is not constructible using the axioms. However, one can describe an initial object as a convex shape and provide examples where one can follow the steps satisfying the axioms to test a theorem. Since it is not possible to draw a general convex shape, prior to the invention of formal logic, reasoning about such initial objects was limited to use of *Syllogism*, which is basically natural language retracted with some constraints and a set of rules that describe valid inference.

The axioms in Euclid's *Elements* are limited to constructions of points, lines and circles. Therefore, in a pre-grammar for initial objects, it is sufficient to talk about the relation that a point is on a line, or the relation that a point is on a circle. This is all the information one needs to explain the relevant theorem. When a point `P` is on a line `L`, one can express this as the proposition `P => L`. Similarly, when a point `P` is on a circle `C`, one can express this as the proposition `P => C`.

In the pre-grammar of initial objects, there is no proposition that expresses the idea of a point in itself. Similarly, there is no line in itself, nor a circle in itself. Points, circles and lines only exist in the sense that they are given names. The names are used in relation to each other, to form propositions that encode the information one associated with the state of initial objects.

In Path Semantics, it is possible to state the proposition `P`, if `P` is a point. This is because, in the foundation of Path Semantics, one can treat everything as a proposition. The name of a proposition, e.g. the string "P" when one has a proposition `P`, is treated as a taboo knowledge from within logic. The user of logic is allowed to have this knowledge, but logic itself can not tell what this string might be, because it would lead to symbolic indistinction, a form of propositional infinity. The pre-grammar of initial objects in Euclid's *Elements* is more limited than Path Semantics. Therefore, the relevant theorem might have applications that are not related to geometry as such.

The pre-grammar of initial objects gives a perspective where something comes before a system of construction. Within the system, the axioms determine what can be expressed. There are no such constraints on the initial objects themselves, but the pre-grammar gives relevant information.

It is not possible to use the pre-grammar to do constructions in the way Euclid's *Elements* is doing. The pre-grammar is a way to reason formally about what kind of information is relevant in initial objects to get started, without depending on the particular configuration in some space.

For example, one might need an initial state of two lines that intersect and are perpendicular. The perpendicular property is not possible to express in the pre-grammar only through relations that points are on lines and circles. However, by using a simpler initial state, one can construct two lines that intersect and are perpendicular, using the axioms. This means, the initial state can be reduced by adding steps, where a cutoff determines what counts as a starting point for some constructive argument. While perpendicularity is not expressible in itself, it follows from a series of steps, such that the simpler initial objects only need to deal with relations of points on lines and circles. The underlying space might be curved and distorted, but the property of perpendicularity will hold from within the space, by the constructive steps.

Therefore, in the pre-grammar, one is only concerned about the information that is not possible to obtain through construction using the axioms. The fundamental relation here is that a point is on line or a circle. All other information is irrelevant, because it is either part of the initial objects that do not matter for the sake of an argument, that might change between examples of the same theorem, or, it might be irrelevant, because this information is possible to get through some construction. This idea is why the discovery of irrational numbers were such a big deal in Ancient Greece. Some irrational numbers are desirable as initial objects, but they are not possible to construct using the axioms. One can argue that the pre-grammar should be extended to support rational numbers, but the problem is that there is no upper limit to how much the pre-grammar might be extended. For the sake of the relevant theorem in this paper, one only considers the part of the pre-grammar that is needed to explain the theorem, without making a claim that this is everything that is interesting about the pre-grammar.

What does one mean when one says that a point is on a line? A line might be arbitrary short. Technically, a point can be a line where the end points are the same point. In a curved space, for example, on the surface of a sphere, a line can be straight, which we more accurately describe as a geodesic curve. The equator is one such example. In a curved space, the equator can be straight, yet when you start at one point and move along the equator, perhaps after an infinite distance, you end up back where you started. This might seem counter-intuitive, but this is because the judgement that an equator on the surface of a sphere is not straight, but a circle, comes from comparing the sphere to some surrounding flat Euclidean space. With Euclidean space one means spaces that satisfy four postulates, but in general one can use three of these postulates to reason about curved space. Therefore, the relation that a point is on a line does not give us any information about the line itself that distinguishes it from a circle. A line might be a circle. A circle might be a line. The end points of a line might be the same, but this does not determine whether the line is a point or a circle. One might use an axiom for the pre-grammar that if the two end points of a line is the same, then the line is a point or a circle. Anyway, a such axiom is not relevant to explain the theorem, but this example shows the intuition behind what one means when a point is on a line.

Since a line can be a circle and a circle can be a line, it must be a way to make sense of lines that distinguishes them from circles in some sense. Here, one can use the methods of construction in Euclid's Elements. A line can be drawn between two points. These two points are on the line. On the contrary, when constructing a circle, one chooses a point as the center and one point on the circle. The center might not be on the circle. When a circle's radius is zero or infinite, the center might be thought of as on the circle, without making reasoning inconsistent. It works from an epistemological point of view. The key idea here is that for a line, there are two points on it. For a circle, there is one point of two on the circle. This is the basic principle for the pre-grammar used to explain the relevant theorem. One can think of propositional AND as a line and OR as a circle.

The basic information is to describe a point on a line or a circle, but lines and circles are using propositional AND and OR, respectively, of such relations.

Line: `(P1 => L) & (P2 => L)`
Circle: `(P1 => C) | (P2 => C)`

Of these two forms of initial objects, the circle is the most baffling one. In the eyes of ancient people, the circle is more mystical in nature than the line. Therefore, the mystery cult is intrinsically associated with the inner circle of people that run the cult. Not all mystery cults used logic in this way to reflect analogous truths between one language and another, a mathematical language with a social language, but the general tendency was a belief that the inner group of their own mystery cult explained the harmony of all existence, that could transcend into such analogous truths. If you had a time machine and went back in time to explain this to them, most of them might agree, because the argument highlights their own importance and desire for more power, however, they might object that your argument applies to other mystery cults than their own. Correctness in the language of mystery cults is very subjective, where truths are seen as loyalty commitment to some particular group. Absoluteness of truths is simply the extension of that particular group's desire for more power onto the whole cosmos. This perspective ignores the views of other mystery cults, because they are seen as competing for the same resources. All mystery cults borrow the same ideas from each other, but they all claim to have invented these ideas and produce stories to convince their followers of some historical origin that links these ideas to their own savior figure.

The idea of a property in the ancient world is a geographical region which is thought to be ruled by some deity. Within this region, the desire for power permeates existence, which gets canceled at the boundaries of this region against the force of competing deities, ruling over other regions. Within the region, the deity has a home, symbolized with a temple, which is kind of like a center of a circle. The circle represents the boundary between one metaphysical domain and another, which must be protected using force. The language of mystery cults takes the power structure in society and applies it geographically, to occupy different regions of metaphysical identity. While it might seem obvious to try to expand a region infinitely, without a boundary, this was usually seen as impolite and disrespectful toward other people's views, which belong to different mystery cults. While the desire for more power is there to permeate a region, it happens within a boundary. Again, like how children play in a playground, where the grown ups set up boundaries, both physically and mentally, of what the children are allowed to do.

Therefore, it might seem strange to modern people to read ancient texts, particularly religious texts, where most sects make similar claims, that would be logically inconsistent if they were all true. Instead of being confused by this seemingly inconsistency, it is better to think of it as attempt to manipulate and exploit people who are not smart enough to understand the difference between loyalty and truth. Confusion is the boundary. It is what makes people feel more comfortable within their own customs and rituals. In the ancient world, it is easier for a person to tolerate others if they display symbols that signify their belonging to some group, because they can more easily predict what might happen in near term future, when being around them. When a person enters a neighborhood village and meets another person, they might be hostile to each other, but when these two meet while traveling, they might behave as friends. In a modern mindset, this seems to be a contradiction, because after Kant, philosophers started thinking about consistent behavior across various social contexts. However, in the ancient world, inconsistent sets of behavior were part of the social constructs. If you examine closely how people use symbols today, then you can find examples where such inconsistencies are still used to create boundaries. The difference is that modern people expect that there ought to be no such inconsistencies. As a way to impose their own claim to objective morality onto everybody else. Now, on top of this, there is also a lot of propaganda that such claims to consistency are fake, to remove the space for friends as travelers.

Anyway, when one takes the perspective of the pre-grammar of initial objects, the idea of a line is that there are two points on a line. There might be more points, but two points are minimum. These two points might be the same point, but they count as two with respect to the line, because they are thought of as the end points of the line. Since there are two end points, two end points can be just one point, but from a certain perspective, that one point can be two points.

One can explain this using the following theorem:

(P => L) → (P => L) & (P => L)

The line can be thought of as a statement about two points on it,
but these two statements can be encoded using a single statement if the two points are the same.

Now, if `(P => L1)` and `(P => L2)`, which means that a point `P` are on the lines `L1` and `L2`, these are two different propositions when `L1` and `L2` are not the same line, but they are the same proposition if `L1` and `L2` are the same line.

So, when one has a line from `P1` to `P2` and from `P2` to `P3`, this can be thought of as two lines, which are one, but it also can be compactified into the following statement:

(P1 => L) & (P2 => L) & (P3 => L)

This leads to the general idea of propositional AND, not merely two terms in the pre-grammar for lines as initial objects in geometry. A such compactification is natural and supported by the *Syllogism*. The problem is that there is no corresponding analogue for circles and propositional OR.

Using the idea of propositional AND, one can create a composed initial object from two initial objects. If `A` and `B` are initial objects, then `A & B` is an initial object. This makes it possible to express the following statement of a circle with three points:

((P1 => C) | (P2 => C)) & ((P2 => C) | (P3 => C))

Intuitively, there are two such solutions for a finite non-zero radius. One is that `P2` is on the circle. In that case, one needs just one point on the circle and can use one of the other two points as center. The other solution is that `P1` and `P3` are on the circle. In that case, `P2` is the center.

The focus here is a statement about two circles, in which the two circles are the same circle. However, these two circles also share one point. This point might be the center, or it might be at the circle. This in itself might not seem very mystical, but when you try to compactify, in a similar way as for propositional AND, to get a propositional OR, you get problems:

(P1 => C) | (P2 => C) | (P3 => C)

This is not logically equivalent to the previous statement. It requires that either `P2` is on the circle, or that `P1` and `P3` is on the circle. Now, one might argue that in Euclid’s Elements, a circle is always constructed using two points, so a such equivalence is meaningful. The mystical aspect here is what makes it sensible to have a such equivalence in geometry, but not in the general *Syllogism*.

In metaphysics, this might have been perceived as problem for ancient people, because they had great confidence that Euclid’s Elements provided an account of geometry, which in turn relates to their idealized forms and their social activity, including mystery cults and power structures. To not be able to explain propositional OR, might have been thought of as an unsolved puzzle. A mystery.

Notice that a circle here, is a relation between two points, so when a circle is related to three points, one must describe it as two circles in the pre-grammar for initial objects. However, in the compactification, which is valid within the language of geometry, there are three propositional terms, which become one at the level of propositional OR. This is how the duality of a circle becomes a trinity. A trinity that can be seen as a duality, but at the end of the day, it might be also be thought of as one.

This provides the basis for the argument that the unification of a duality of circles, with a trinity of circles, gives ancient people the motivation to look for something that can convince them to put the duality first, or the trinity first. An argument for whether the *Syllogism* should come before the *Elements*, or the *Elements* before the *Syllogism*. At the outset, a such question might not seem relevant to mystery cults, but as this puzzle gets wide spread, solving it, or claiming to, might attract followers to particular mystery cults. Again, because mystery cults keep their ritual of communion with the divine a secret, they can make such claims without needing to give a proof. People might have been inclined to believe they had solved the puzzle, as evidence of their own views being correct, without actually having a satisfactory analytic solution. While the puzzle itself might be hard to solve, people who lacked the skills to solve it might like the idea of being able to solve it. They might pretend to, or convince themselves they did it, because other people make similar claims and falling behind was perceived as a threat to their own social identity.

A such friendly competition might also seem childish to some ancient people, who thought of it as something of interest to philosophers, but not of relevance to normal religious practice. Which means, most people simply continued with a polytheistic mindset, while a few fanatics obsessed over whether there might be one original deity, two or three. It requires another level of cultural anxiety, to explain how a such idea comes at the forefront of theological debate. This happens because after the 1st and 2nd wars between the Roman empire and Judea, some Jews are starting to seriously think about their own religious views and the way forward. In the 40s of the 2nd century, Marcion of Sinope suggests that the deity that Jews worships was evil, but there was a secret savior figure, Jesus Christ, that one can find in the old texts, if one looks closely. Marcion is not the first person who does this, but the first to canonize the texts he presented to the world. By default, a deity must have a home, so when Jews loses the access to the temple, their deity loses its home and accordingly, must be looking for one. This means, they start to search for a unifying theology to explain away the fact that their deity has no concrete home at the moment, but will have soon. However, to create that home, it requires the Roman empire to change, which means this cultural anxiety is expressed as a belief in the end of the world. The end of the Roman empire. Apocalypse.

The Early Christians in the 2nd century have the precise cultural anxiety that makes it possible to put a metaphysical problem in geometry at the forefront of theological debate between mystery cults. While none of these people seem to solve the problem, they are under influence of ideas that come from philosophy. How Christianity ended up with a Holy Trinity, is more or less a historical accident, but to explain how it became seen as important, one needs both a way to spread a puzzle that might inspire a debate of divine trinity versus a divine duality, plus a cultural anxiety that drives whole groups of people to identify themselves with such theology. In most polytheistic mythology, the emphasis is on the home of the deity. In Christian mythology, the emphasis is still on a home of a deity, but it is not a home that exists. The home is lacking and will to come at the end of time.

The relevant theorem is based of erasure of the specific statements of the form that a point is on a circle. Because, it is unnecessary, as long one is talking about equality of propositions.

For example, if `((P1 => C) | (P2 => C)) & ((P2 => C) | (P3 => C))`,
then one can replace `(P2 => C)` with `a`: `((P1 => C) | a) & (a | (P3 => C))`.
When doing this to all terms, one gets: `(b | a) & (a | c)`.

To guard the statement during compactification, one can use an exponential proposition:

$$!!a \rightarrow (b \mid a \mid c) == ((b \mid a) \,\&\, (a \mid c))$$

This choice is simply one of the solutions, that there is one point on the circle. The other case where two points are on the circle, might be logically equivalent when this holds for all propositions. Further research needs to be made to determine which such expressions belong to the same class of ideas. However, the main argument here is that, once there is a such idea, one can prove:

$$true \rightarrow all(excm(a)\hat{}true) == all(!!a \rightarrow (b \mid a \mid c) == ((b \mid a) \,\&\, (a \mid c)))$$

With other words, the puzzle of the unification with a divine trinity and a divine duality, is precisely the statement of the law of excluded middle, but this proof only works with universals.

The relevant theorem provides an account for seeing this puzzle, in the tension between the *Elements* and the *Syllogism*, not as a problem that needs to be solved, but itself a potential justification for both the law of excluded middle and universals, in the view of ancient philosophers.

This theorem, or similar ideas, might be at the core of western philosophical thought.

Q.E.D.

Appendix A

Proof of the relevant theorem in Hooo:

```
fn proof_1 : excm(a)^true -> (!!a -> (b | a | c) == ((b | a) & (a | c))) {
    use std::tauto_imply_to_pow;

    x : excm(a)^true;

    fn f : excm(a) -> !!a => ((b | a | c) == ((b | a) & (a | c))) {
        x : excm(a);

        lam r : !!a => ((b | a | c) == ((b | a) & (a | c))) {
            use std::refl;
            use std::right;
            use std::left;

            y : !!a;

            let g1 = refl() : a => a;
            lam g2: !a => a {
                z : !a;
                let z2 = y(z) : false;
                let r = match z2 : a;
                return r;
            }
            let y2 = match x (g1, g2) : a;
            lam f1 : (b | a | c) => ((b | a) & (a | c)) {
                z : b | a | c;
                let g1 = right(y2) : b | a;
                let g2 = left(y2) : a | c;
                let r = refl(g1, g2) : (b | a) & (a | c);
                return r;
            }
            lam f2 : ((b | a) & (a | c)) => (b | a | c) {
                z : (b | a) & (a | c);

                let g1 = left(y2) : a | c;
                let r = right(g1) : b | a | c;
                return r;
            }

            let r = refl(f1, f2) : (b | a | c) == ((b | a) & (a | c));
            return r;
        }
        return r;
    }
    fn g : b^a -> a^true => b^true {
        x : b^a;
        lam r : a^true => b^true {
            use std::pow_transitivity;

            y : a^true;
            let r = pow_transitivity(y, x) : b^true;
            return r;
        }
        return r;
    }

    let f2 = g(f) :
        excm(a)^true => (!!a => ((b | a | c) == ((b | a) & (a | c))))^true;
    let x2 = f2(x) : (!!a => ((b | a | c) == ((b | a) & (a | c))))^true;

    let r = tauto_imply_to_pow(x2) : !!a -> (b | a | c) == ((b | a) & (a | c));
    return r;
}
```

The first lemma, `proof_1`, establishes that from excluded middle, one can get the compactification. This proof does not rely on universals, because it proves it for the specific case of excluded middle for one proposition. With other words, this lemma does not impose the universal excluded middle as a metaphysical concept. The excluded middle is a choice, but not necessary.

The meta-strength of `proof_1` is 2, according to the HOOO EP 2026 Standard. It means, that this proof is accepted by most path semanticists as valid under reasonable and common assumptions of meta-theorem proving. However, just because this proof is accepted, does not mean it makes ontological or metaphysical claims. It is simply a way to measure the proof relative to normal constructive logic. Since the theorem uses exponential propositions, it is not possible to prove this with normal constructive logic. Meta-strength 2 is a reasonable level for meta-theorem proving, which is common when dealing with theorems using exponential propositions.

```
fn proof_2 : all(!!a -> (b | a | c) == ((b | a) & (a | c))) -> all(excm(a)^true) {
    use std::pow_transitivity;
    use std::pow_to_tauto_imply;

    x : all(!!a -> (b | a | c) == ((b | a) & (a | c)));

    let x2 = x() : all(!!a -> (false | a | true) == ((false | a) & (a | true)));
    fn f : (false | a | true) == ((false | a) & (a | true)) -> a {
        use std::fst;
        use std::eq_transitivity;
        use std::refl;
        use std::imply_lift;
        use std::right;
        use std::absurd;
        use std::left;

        x : (false | a | true) == ((false | a) & (a | true));

        let tr = () : true;
        lam f1 : true => (false | a | true) {
            y : true;

            let y2 = right(y) : a | true;
            let r = right(y2) : false | a | true;
            return r;
        }
        let f2 = imply_lift(tr) : (false | a | true) => true;
        let x2 = refl(f1, f2) : true == (false | a | true);

        lam g1 : ((false | a) & (a | true)) => a {
            y : (false | a) & (a | true);
            let y2 = fst(y) : false | a;
            let f1 = absurd() : false => a;
            let f2 = refl() : a => a;
            let r = match y2 (f1, f2) : a;
            return r;
        }
        lam g2 : a => ((false | a) & (a | true)) {
            x : a;
            let f1 = right(x) : false | a;
            let f2 = left(x) : a | true;
            let r = refl(f1, f2) : (false | a) & (a | true);
            return r;
        }
        let x3 = refl(g1, g2) : ((false | a) & (a | true)) == a;

        let x4 = eq_transitivity(x2, x) : true == ((false | a) & (a | true));
        let x5 = eq_transitivity(x4, x3) : true == a;
        let x6 = fst(x5) : true => a;

        let r = x6(tr) : a;
        return r;
    }

    let yf = f() :
        all((false | a | true) == ((false | a) & (a | true)) -> a);
    let x3 = pow_transitivity(x2, yf) : all(!!a -> a);

    fn f2 : all(!!a => a) -> all(excm(a)) {
        use std::nnexcm;
        use std::triv;

        x : all(!!a => a);

        let y = x() : all(!!excm(a) => excm(a));
        let y2 = nnexcm() : true -> all(!!excm(a));
        let x2 = triv(y2) : all(!!excm(a));
        let x3 = y(x2) : all(excm(a));
        let r = x3() : all(excm(a));
        unsafe return r;
    }

    let x4 = pow_to_tauto_imply(x3) : all(!!a => a)^true;
    let r = pow_transitivity(x4, f2) : all(excm(a)^true);
    unsafe return r;
}
```

The second lemma, `proof_2`, establishes that from the compactification, one can get the law of excluded middle. This proof relies on universals. The `unsafe return` statements are due to the safety heuristics in Hooo, because returning a universal is often not what the user intended. It serves as warning, but does not make the deduction invalid. One can think of this lemma as somebody who likes the *Elements* and wants to argue for the law of excluded middle in the *Syllogism* in general. A such position might be from the perspective of attributing Platonic ideal forms as a greater insight into the nature of existence, in according to what mystery cults perceived communion with the divine as the basic explanation for existence and human experience. The lemma supports the view that if one starts with the perspective of mystery cults in the ancient world and works backwards toward a system for reasoning, it seems to make sense that there exist universals metaphysically. However, this is based on the assumption that there is something more important about the *Elements* that gives it metaphysical priority, over a more general Syllogistic framework of thought.

The meta-strength of `proof_2` is 2, according to the HOOO EP 2026 Standard. It means, that this proof is accepted by most path semanticists as valid under reasonable and common assumptions of meta-theorem proving. However, just because this proof is accepted, does not mean it makes ontological or metaphysical claims. It is simply a way to measure the proof relative to normal constructive logic. Since the theorem uses exponential propositions, it is not possible to prove this with normal constructive logic. Meta-strength 2 is a reasonable level for meta-theorem proving, which is common when dealing with theorems using exponential propositions.

```
fn proof_3 : true -> all(excm(a)^true) == all(!!a -> (b | a | c) == ((b | a) & (a | c))) {
    use proof_1;
    use proof_2;
    use std::refl;

    x : true;

    let f1 = proof_1() :
        all(excm(a)^true) -> all(!!a -> (b | a | c) == ((b | a) & (a | c)));
    let f2 = proof_2() :
        all(!!a -> (b | a | c) == ((b | a) & (a | c))) -> all(excm(a)^true);
    let r = refl(f1, f2) :
        all(excm(a)^true) == all(!!a -> (b | a | c) == ((b | a) & (a | c)));
    unsafe return r;
}
```

The third lemma, `proof_3`, brings together the first lemma, `proof_1` and the second lemma `proof_2`, to prove the entire relevant theorem. Because the second lemma relies on universals, the entire relevant theorem relies on universals. There is nothing about this theorem that gives excluded middle, nor the compactification of circles to propositional OR, some ontological or metaphysical priority. However, it supports the argument that under influence of Euclid's *Elements*, this might have motivated a Seshatic-Platonic language bias cycle that changes how Early Christians think about theology during the 2nd century. These ideas are probably not invented by Early Christians themselves, but coming from a broader tradition of philosophical thought and practice of medicine.

The meta-strength of `proof_3` is 2, according to the HOOO EP 2026 Standard. It means, that this proof is accepted by most path semanticists as valid under reasonable and common assumptions of meta-theorem proving. However, just because this proof is accepted, does not mean it makes ontological or metaphysical claims. It is simply a way to measure the proof relative to normal constructive logic. Since the theorem uses exponential propositions, it is not possible to prove this with normal constructive logic. Meta-strength 2 is a reasonable level for meta-theorem proving, which is common when dealing with theorems using exponential propositions.